

BLOG

The Holy Grail of Investing

Yicheng Yang · April 18, 2026 · 12 min read

“Put 10% of the cash in short-term government bonds and 90% in a very low-cost S&P 500 index fund. I believe the trust's long-term results from this policy will be superior to those attained by most investors.”

— Warren Buffett, 2013 letter to shareholders

Buffett is right. He's been right for fifty years. For the vast majority of people, the honest answer to “what should I invest in?” is *a low-cost index fund, and then stop looking at it.*

But if you are reading this blog, you probably don't want to stop looking at it. You want to know if there is a simple rule — not day trading, not options, not machine learning — that beats the index over decades. There is. It's been in public literature since 2016. It compounds \$10,000 into \$543,000 over 27 years, versus Buffett's \$150,000. This post explains exactly how it works, what it costs you, and where it fails.

\$543,387

what \$10,000 became in this strategy from 1999 to 2026 — 3.6× the QQQ index, 38× TQQQ buy-and-hold

The rule

Two ETFs. One signal. No discretion.

THE STRATEGY

Every day, check two numbers on QQQ:

- the 5-day exponential moving average (fast trend)
- the 200-day exponential moving average (long trend)

If the fast trend is ABOVE the long trend:

→ hold 100% TQQQ (3× leveraged Nasdaq)

If the fast trend is BELOW the long trend:

→ hold 100% QQQ (unleveraged Nasdaq)

That's it.

The whole point is this: TQQQ in a rising market is magic, but TQQQ in a falling market is ruin. The 200-day moving average is a crude but reliable way to tell which regime you're in. When the trend is up, take the leverage. When the trend breaks, step down to plain QQQ and wait.

This is not my invention. Meb Faber wrote about single-moving-average rules in 2007. Gayed & Bilello won the 2016 Charles H. Dow Award for essentially this idea on the S&P 500. Logan Kane popularized it for TQQQ on Seeking Alpha in 2018. HEDGEFUNDIE's Bogleheads thread has been debating variations for five years. What I'm doing here is implementing it carefully, testing it honestly, and showing you the results without the usual “47% annualized!!!” marketing.

What it did

Starting with \$10,000 in 1999, invested according to this rule through every recession, every bubble, every war, every pandemic:

STRATEGY	ANNUAL RETURN	WORST DRAWDOWN	\$10K FINAL
The rotation (this strategy)	15.88%	-95.5%	\$543,387
QQQ buy & hold	10.52%	-83.0%	\$150,523
S&P 500 buy & hold	8.4%	-55.2%	\$89,000
TQQQ buy & hold (no rule)	1.37%	-99.98%	\$14,439

Look at the last row first. TQQQ unfiltered is not a strategy, it's a way to turn \$10,000 into \$14,000 over 27 years while losing 99.98% along the way. The leverage multiplies the bad years so badly that the good years cannot dig you out.

Now look at the first row. Apply the moving-average rule to the same instrument and \$10,000 becomes \$543,387. The rule does not change which stocks you own — you're still in the Nasdaq-100 the whole time — it just decides when to turn the leverage on and off.

Rotation Strategy vs Buy & Hold — Full History + Regime Zoom

Full History (2000-2026): \$10K Compounded, Log Scale



Dot-com Unwind (2000-2003)



Financial Crisis (2007-2010)



COVID Shock (2020-2020)



AI Bull + 2022 Selloff (2022-2026)



The full 27-year growth chart on a log scale (top), plus four close-ups of the crisis periods. In each crisis window, the rotation (red) is compared to plain QQQ and plain TQQQ. You can see the rotation survive 2000–2003 while TQQQ goes to \$58; you can see it take off after 2017 as the signal stays bullish through the AI rally.

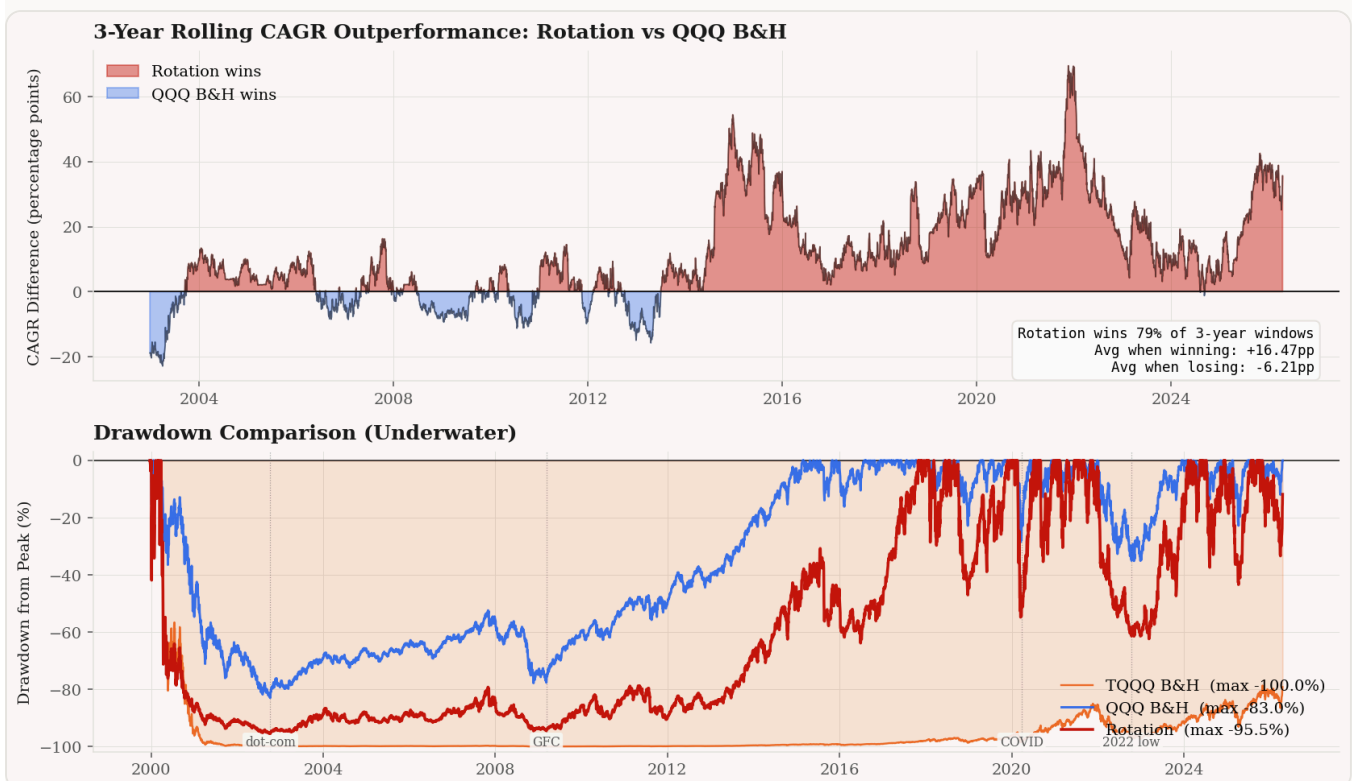
Why it works

Three forces create the edge.

Volatility decay. A 3× leveraged ETF does not deliver 3× the underlying's long-term return. Because it rebalances daily, wild swings compound against you. In a choppy sideways market, TQQQ can lose money while QQQ goes nowhere. The moving-average filter catches most of those choppy periods and switches you out. You only hold the leverage in trending markets, where daily rebalancing helps rather than hurts.

The bear leg still earns. When the trend breaks and you move to plain QQQ, you're still invested in one of the best-performing equity indices in history. That's the key difference from the classic version of this strategy, which sits in T-bills during bear signals. T-bills earn almost nothing in most of the post-2008 period. QQQ earns the equity risk premium. Over 27 years, that adds up.

You stay in the game. Pure TQQQ buy-and-hold would have destroyed you in 2000 and 2008. The rotation didn't. Once your account survives a crisis, you get to participate in the recovery with full leverage turned back on.



Top chart: for every 3-year window since 2003, how much did the rotation beat QQQ buy-and-hold? Red means "the rotation won," blue means "QQQ won." The rotation wins 79% of the time, by an average of 16 percentage points. It loses 21% of the time, by an average of 6 percentage points. Bottom chart: the drawdowns. TQQQ alone went to -100%. QQQ alone went to -83%. The rotation's worst was -95% — brutal, but recoverable.

What it costs you

The big number is a -95% drawdown. That's real. It happened in 2002 during the synthetic TQQQ period, and something similar would happen if we got another dot-com-scale event tomorrow.

Imagine your \$150,000 portfolio becoming \$7,500 on paper. That's not a number you can talk yourself into holding. Most people would sell at \$30,000, right before the recovery. That's how this strategy actually kills people — not because it fails mathematically, but because its worst moments are psychologically unbearable.

So here's the honest deal. This strategy is mathematically better than QQQ buy-and-hold over long horizons. It is also behaviorally worse. If you can't genuinely tell yourself “I would hold through a 95% drawdown,” don't use it. Buffett's 90/10 advice wins for you, and he's right.

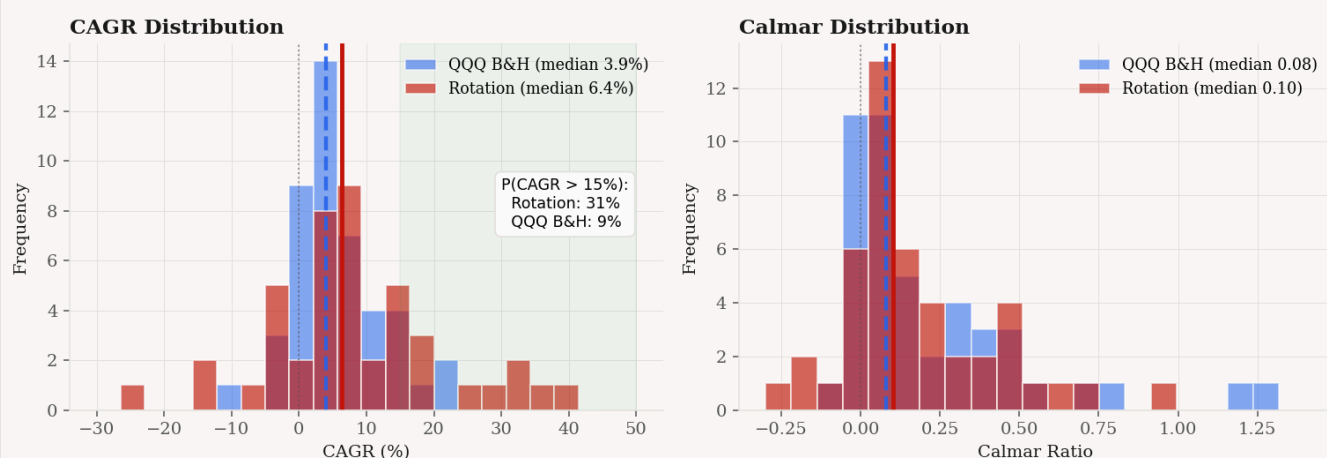
How believable is the 15.88%?

Not as believable as it looks.

I ran this strategy through a stress test called combinatorial purged cross-validation — a fancy name for “rearrange the history 45 different ways and see what happens.” The single historical path gives 15.88%. The median across all 45 rearrangements gives 6.38%. That means the headline number depends heavily on the specific sequence of bull and bear markets we actually got. In other parallel-universe sequences with the same statistics, the strategy did less well.

A more honest expectation: this strategy probably delivers 4–16% annualized in the next 27 years, with a 31% chance of beating 15%. QQQ buy-and-hold in the same test delivers 1–9% annualized with a 9% chance of beating 15%. The rotation is about 3.5 times more likely to deliver a great outcome — but also more likely to deliver a merely-okay outcome, and is still subject to a nightmare drawdown.

CPCV Distributions Across 45 Combinatorial Paths



If you ran the same strategy through 45 slightly different histories, how did it perform? Red is the rotation, blue is QQQ buy-and-hold. The rotation's distribution is shifted to the right — more likely to deliver a good outcome, especially in the right tail above 15% annual returns.

A few things I'd change the framing on

Writing this post honestly means admitting several things.

The 200-day moving average and the 5-day fast signal are a reasonable choice, but they are not uniquely magical. I tested 24 different parameter combinations — both simple and exponential moving averages, with fast windows of 5, 10, 20, 50 days and slow windows of 150, 200, 250 days. EMA(5)/EMA(200) ranks sixth out of 24 on annual return. Most of the other parameterizations also beat QQQ buy-and-hold. The signal family is what matters; the specific numbers are second-order.

Whether you hold QQQ or T-bills in the bear leg turns out to matter less than I'd hoped. The T-bills version (originally from Kane) delivers essentially the same Sharpe ratio and a slightly better Calmar over the full 27 years. The QQQ version's edge shows up mainly in post-2010 data. Either implementation works.

Almost all of the annual-return advantage over QQQ buy-and-hold comes from post-2010 compounding in a period when the Nasdaq 100 was on a historic run. If we re-run starting from 2003, the advantage is still there but smaller. Starting from 2010 (when TQQQ actually launched, so this is the only truly out-of-sample window), the advantage is 16 percentage points per year.

None of this changes the basic claim: the strategy has an edge, it replicates widely in the literature, and it is almost certainly worth taking for an investor with the risk tolerance to withstand it. But it's not magic, and the exact size of the edge is uncertain within a wide range.

How to actually do it

If you want to run this yourself, here's the minimum viable procedure:

Open a brokerage account that lets you trade QQQ and TQQQ — Interactive Brokers, Fidelity, Schwab, Robinhood, almost any U.S. broker. This strategy is best run in a tax-advantaged account (IRA, Roth IRA, 401k where available) because the rotations are taxable events in a regular account; in a taxable account you lose 1–3 percentage points of annual return to short-term capital gains tax.

Every day after market close, compute two numbers: the 5-day and 200-day exponential moving averages of QQQ's closing price. If the 5-day is above the 200-day, your target position is 100% TQQQ. Otherwise, your target is 100% QQQ. If your current position doesn't match the target, rebalance the next trading day.

Expect to rotate about three to four times per year. At 2–3 bps of total trading cost per rotation, friction is negligible.

Size the position to survive a 95% drawdown without panic-selling. If this strategy would wipe out your emergency fund, put less money in. I personally run this on ~30% of my investable capital, with the rest in lower-risk allocations. The expected return is lower that way, but the expected value of “me sticking with the plan” is much higher.

Interactive equity curves available at yichengyang-ethan.github.io/holy-grail

The bottom line

Buffett's advice is the right answer for most people. If you don't want to think about markets, if you don't want to see your portfolio cut in half every decade, if you want to spend your attention on things other than money — take his advice and buy VOO or VTI and ignore it forever.

But if you've decided you want to be more active than that, and you've decided you can handle leverage, and you understand that the pure TQQQ version of this bet has historically ended in ruin — then the moving-average rotation is the simplest thing I can think of that has a defensible statistical edge and an implementation that fits on a napkin. It's been known in the literature for a decade. It's not new. It's not secret. It is, as close as anything retail investors have, a Holy Grail.

Within the limits of what we can prove with 27 years of history and one stress-tested distribution.